NIKOLAOS ROUFAS

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LinkedIn | Website | GitHub | Google Scholar | ORCID

Research Interests: Machine Learning · Deep Learning · Explainable AI · Natural Language Processing · Retrieval-Augmented Generation

Computer Science undergraduate and AI researcher with **4 peer-reviewed publications and 2 manuscripts under review**, including multiple first-author works spanning natural language processing, legal AI, transformer explainability, scientific machine learning, and quantum machine learning. Research experience includes the design, implementation, and evaluation of transformer-based and intrinsically interpretable models, with international research experience at Sapienza University of Rome.

EDUCATION

Ionian University — B.Sc. in Computer Science **Sep. 2024 – Jun. 2028 (Expected)**
Department of Informatics *Corfu, Greece*

- GPA: **7.92/10**; approximately **top 3%** of a cohort of ∼150 students.

Sapienza University of Rome — Erasmus+ Exchange Student **Sep. 2025 – Jul. 2026**
Department of Computer, Control and Management Engineering Antonio Ruberti (DIAG) *Rome, Italy*

- Conducted research within the Processes, Services and Software Management (PSSM) Lab under Prof. Massimo Mecella.

INDUSTRY EXPERIENCE

Artificial Intelligence Researcher (Contract) **Jun. 2025 – Oct. 2025**
Stealth Startup

- Researched and evaluated **10+ large language and multimodal models** across performance, robustness, and alignment dimensions for R&D and pre-launch validation.
- Independently built Python/PyTorch evaluation tooling that automated most of the benchmarking workflow, reducing manual evaluation effort and improving experimental reproducibility.
- Collaborated within a **5 person technical team**, reporting directly to the technical lead while supporting model experimentation, validation, and internal research under NDA.

RESEARCH EXPERIENCE

Undergraduate Researcher **Nov. 2024 – Present**
Humanistic and Social Informatics Lab (HILab), Ionian University — Prof. Andreas Kanavos *Corfu, Greece*

- Developed **LegNER**, a domain-adapted BERT model for legal named-entity recognition and anonymization, achieving **99.4% F1** on a corpus of **1,542 court cases** by combining legal-domain pretraining, span-level supervision, and an expanded legal vocabulary.
- Developed and benchmarked transformer-based sentiment classifiers for climate-change discourse, achieving **90% F1** with RoBERTa on a curated **1,000-post Reddit dataset** using a multi-stage annotation and human-validation pipeline.
- Investigated layer-wise explainability in BERT across **12 transformer layers**, identifying a **56% increase in Local Intrinsic Dimensionality** and a strong association with SHAP attribution strength ($r = 0.77$**,** $p < 0.01$).
- Developed a Performer-based protein-structure prediction model combining linear attention with PSSM evolutionary features, demonstrating **3–5× lower memory use** and **2–4× faster training** for long protein sequences.
- Implemented and evaluated **variational quantum classifiers** across multiple feature-map/ansatz configurations, using gradient-norm and gradient-variance analysis to study the **expressivity–trainability trade-off** and barren-plateau behavior.
- Maintain the lab's Linux research server with **2× NVIDIA RTX 5000 GPUs** supporting five faculty/researchers, including CUDA/PyTorch environments, Docker, user access, storage, backups, monitoring, troubleshooting, and job scheduling.

Visiting Research Student **Sep. 2025 – Jul. 2026**
PSSM Lab, DIAG, Sapienza University of Rome — Prof. Massimo Mecella *Rome, Italy*

- Led end-to-end development of **LegalSENN**, an intrinsically interpretable transformer architecture for legal document classification, combining Legal-BERT representations with a concept bottleneck, input-dependent relevance scores, and additive prediction.
- Evaluated the architecture on **7,800 SCOTUS opinions across 13 legal issue classes** over five random seeds, reaching **0.639 macro-F1** while preserving intrinsic, forward-pass explanations.
- Demonstrated explanation faithfulness through concept-masking experiments: removing the six highest-relevance concepts caused a **41.5% label-flip rate**, compared with roughly **10%** for random or least-important concepts.
- First author of the resulting manuscript, currently under review at *Discover Artificial Intelligence*; developed and maintain the accompanying public research codebase on GitHub.

TEACHING EXPERIENCE

Peer Tutor (Informal) — Ionian University

- Supported **7 fellow students** through informal tutoring and coursework guidance in Operating Systems, Artificial Intelligence, Data Mining, and Databases.

PUBLICATIONS & MANUSCRIPTS

- **Roufas, N.**, Marinacci, M., Kanavos, A., & Mecella, M. "Self-Explaining Transformer Architectures for Legal Document Classification." *Under review at Discover Artificial Intelligence*, Springer Nature, 2026.
- Karamitsos, I., **Roufas, N.**, & Kanavos, A. "Expressivity, Trainability, and Gradient Scaling in Variational Quantum Classifiers." *Under review at Discover Computing*, Springer Nature, 2026.
- Karamitsos, I., **Roufas, N.**, Al-Hussaeni, K., & Kanavos, A. "LegNER: A Domain-Adapted Transformer for Legal Named Entity Recognition and Text Anonymization." *Frontiers in Artificial Intelligence*, 8, 1638971, 2025.
- **Roufas, N.**, Mohasseb, A., Karamitsos, I., & Kanavos, A. "Analyzing Public Discourse and Sentiment in Climate Change Discussions Using Transformer-Based Models." *IFIP International Conference on Artificial Intelligence Applications and Innovations (AIAI)*, pp. 39–53, Springer, 2025.
- **Roufas, N.**, Kanavos, A., Karamitsos, I., Al-Hussaeni, K., & Maragoudakis, M. "Do Deeper Layers Explain Better? An LID-Based Study of Transformer Explainability." *2025 IEEE International Conference on Big Data (BigData)*, pp. 6552–6559, 2025.
- **Roufas, N.**, Karamitsos, I., Al-Hussaeni, K., Gerogiannis, V. C., & Kanavos, A. "Efficient Protein Folding with Transformer Models Using the Performer Architecture." *International Conference on Advances in Information and Communication Technology*, 2025.

HONOURS, AWARDS & FUNDING

- **Erasmus+ Mobility Grant** — **€2,000**, awarded through the Ionian University Erasmus+ Office to support mobility at Sapienza University of Rome, 2025–2026.
- **Academic Excellence Award** — First High School of Florina, 2024; awarded for achieving a final high-school average of **19/20**.
- **4th Place, Hellenic Fencing Championships — Men's Team Épée**, Hellenic Fencing Federation, 2024.
- **Creativity Award / National Finalist, WRO Hellas Robotics Competition**, 2017; won the Western Macedonia regional competition with a LEGO Earth–Moon space-vehicle concept and placed **8th nationally among 1,343 teams**.

LEADERSHIP & SERVICE

IT, Logistics & PR Coordination Team **Feb. 2025 – Present**
Ionian University Student Conference in Informatics *Corfu, Greece*

- Coordinated communication, logistics, and event operations as part of a **20-member organizing team** for a university conference with approximately **200 attendees and 20 speakers**.

Erasmus+ Buddy **Feb. 2025 – Jul. 2025**
ESN Ionian *Corfu, Greece*

- Supported **3 incoming Erasmus students** with practical guidance, cultural integration, and participation in local student activities.

TECHNICAL SKILLS

Programming & Scientific Python	Python, NumPy, pandas
Machine Learning & Deep Learning	PyTorch, TensorFlow, scikit-learn, Keras, Hugging Face Transformers
NLP, Vision & Quantum ML	NLTK, OpenCV, Qiskit
Research Infrastructure	Linux, Git/GitHub, Docker, Kubernetes, CUDA/GPU environments, Jupyter
Mathematics & Scientific Computing	Linear Algebra, Probability, Statistics, Optimization, Numerical Methods
Visualization	Matplotlib

LANGUAGES

Greek (Native) · English (C2) · French (B1) · Portuguese (A2)